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Smoothing the Lee–Carter and Poisson log-bilinear models for mortality forecasting: a penalized log-likelihood approach

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Abstract: Mortality improvements pose a challenge for the planning of public retirement systems as well as for the private life annuities business. For public policy, as well as for the management of financial institutions, it is important to forecast future mortality rates. Standard models for mortality forecasting assume that the force of mortality at age x in calendar year t is of the form $\exp(\alpha_x + \beta_x\kappa_t)$. The log of the time series of age-specific death rates is thus expressed as the sum of an age-specific component α_x that is independent of time and another component that is the product of a time-varying parameter κ_t reflecting the general level of mortality, and an age-specific component β_x that represents how rapidly or slowly mortality at each age varies when the general level of mortality changes. This model is fitted to historical data. The resulting estimated κ_t's are then modeled and projected as a stochastic time series using standard Box–Jenkins methods. However, the estimated β_x's exhibit an irregular pattern in most cases, and this produces irregular projected life tables. This article demonstrates that it is possible to smooth the estimated β_x's in the Lee–Carter and Poisson log-bilinear models for mortality projection. To this end, a penalized least-squares/maximum likelihood analysis is performed. The optimal value of the smoothing parameter is selected with the help of cross validation.

Key words: cross validation; life table; mortality projection; roughness penalty; smoothing

Data and software link available from: http://stat.uibk.ac.at/SMIJ

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1 Introduction

In most industrialized countries, mortality at adult and old ages reveals decreasing annual death probabilities. Mortality improvements pose a challenge for the planning of public retirement systems as well as for the private life annuities business. More generally, all the components of social security systems, including disability and survivorship benefits, as well as medical care for the aged, are affected by mortality trends. When long-term living benefits are concerned, the calculation of

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expected present values (for insurance pricing or reserving) requires an appropriate mortality projection in order to avoid underestimation of future costs. Traditional life insurance and pension plan valuation thus relies on the accurate modeling of future mortality. Over the years a number of actuaries tried to get better, more complex formulas to provide a fit to more of the age range. Nowadays, modern actuaries are less interested in these analytical 'laws' of mortality, and instead prefer simply using smoothed mortality experience.

We analyze the changes in mortality as a function of both age x and calendar time t. Henceforth,

- $T_x(t)$ is the remaining lifetime of an individual aged x on January the first of year t; this individual will die at age $x + T_x(t)$ in year $t + T_x(t)$.
- $q_x(t)$ is the probability that an x-aged individual in calendar year t dies before reaching age $x + 1$, i.e. $q_x(t) = \Pr[T_x(t) \leq 1]$.
- $p_x(t) = 1 - q_x(t)$ is the probability that an x-aged individual in calendar year t reaches age $x + 1$, i.e. $p_x(t) = \Pr[T_x(t) > 1]$.
- $\mu_x(t)$ is the mortality force at age x during calendar year t, that is,

$$\mu_x(t) = \lim_{\Delta \to 0} \frac{\Pr[x < T_0(t-x) \leq x + \Delta | T_0(t-x) > x]}{\Delta}.$$

Lee and Carter (1992) proposed a simple model for describing the secular change in mortality as a function of a single time index. Throughout this paper, we assume that the age-specific mortality rates are constant within bands of age and time, but allowed to vary from one band to the next. Specifically, given any integer age x and calendar year t, the identity

$$\mu_{x+\xi}(t + \tau) = \mu_x(t) \text{ for } 0 \leq \xi, \tau < 1 \tag{1.1}$$

holds true. The Lee–Carter model specifies a log-bilinear form for the force of mortality $\mu_x(t)$, that is

$$\ln \mu_x(t) = \alpha_x + \beta_x \kappa_t. \tag{1.2}$$

Interpretation of the parameters is quite simple: $\exp \alpha_x$ is the general shape of the mortality schedule and the actual forces of mortality change according to an overall mortality index κ_t modulated by an age response β_x (the shape of the β_x profile tells which rates decline rapidly and which slowly over time in response of change in κ_t). This model is fit to historical data and the resulting estimate of the time-varying parameter is then forecast as a stochastic time series using standard Box–Jenkins methods. From this forecast of the general level of mortality, the actual age-specific rates are derived using the estimated age effects. For a review of recent applications of the Lee–Carter methodology, we refer the interested reader to Lee (2000).

The main statistical tool of Lee and Carter (1992) is least-squares estimation. This implicitly means that the errors are assumed to be homoskedastic, which is quite unrealistic: the logarithm of the observed force of mortality is much more variable at older ages than at younger ages because of the much smaller absolute number of deaths at older ages. Various alternative esitmation procedures have

been compared in Wilmoth (1993), who proposed a Poisson model for Lee–Carter rates. In the actuarial literature, Sithole *et al.* (2000) and Renshaw and Haberman (2003a, 2003b) have implemented an alternative approach to mortality forecasting based on heteroskedastic Poisson error structures. A closely related model has been proposed by Brouhns *et al.* (2002a, 2002b), keeping the Lee–Carter log-bilinear form for the forces of mortality but replacing ordinary least-squares regression with Poisson regression for the death counts. There is thus a key difference between Renshaw and Haberman (2003b) and the method proposed by Brouhns *et al.* (2002a, 2002b): the difference centres on the interpretation of time, which in the Lee–Carter and Brouhns *et al.* (2002a, 2002b) approach is modelled as a factor and under the approach proposed by Renshaw and Haberman (2003b) is modelled as a known covariate (see also the last article for a comparison in the Lee–Carter case).

In empirical studies, the estimated β_x's exhibit an irregular pattern. This may be regarded as undesirable from an actuarial point of view, since the resulting projected life tables will also show some erratic variations across ages. Actuaries use these life tables to compute life insurance premiums as well as reserves that have to be held by insurance companies to make them able to pay the future contractual benefits. Irregularities in the life tables then propagate to the price list and to balance sheets, which is not desirable. Therefore, as long as these irregularities do not reveal particular features of the risk covered by the insurer, but are likely to be caused by sampling errors, actuaries prefer to resort to statistical techniques producing regular life tables. As pointed out by Loader (1999), smoothing methods and local regression originated in actuarial science in the late nineteenth and early twentieth centuries, in the problem of graduation. Given a set of mortality rates, the actuary is asked to adjust the observations so that the graduated values of the series capture the main trend in the data. Early references include Spencer (1904), Sheppard (1914a, 1914b) and Henderson (1916). See the review by Seal (1981). Smoothing splines and penalized likelihood methods also originate in actuarial science after Whittaker (1923) and Henderson (1924).

A naive solution consists in smoothing the estimated β_x's, without modifying the α_x's or the κ_t's. This may however produce large discrepancies between observations and model predictions. To avoid this problem, the smoothing of the β_x's could be performed in the estimation phase. A convenient way to impose some smoothness is to introduce a roughness penalty via penalized least squares in the Lee–Carter model and penalized maximum likelihood in the Poisson log-bilinear model. We will come back to the comparison of the ad hoc smoothing of the estimated β_x's with penalized least squares or likelihood estimators in the conclusion to this article.

Note that the estimated α_x's are usually very smooth, since they represent an average effect of mortality at age x over the observation period. The estimated κ_t's are often rather irregular, but the projected κ_t's, obtained from some time series model (like ARIMA), will be smooth. Hence, we only need to smooth the β_x's in order to get projected life tables, with mortality varying smoothly accross the ages. Let us mention that Renshaw and Haberman (2003c) suggested to smooth the estimated β_x's. This article aims to implement this idea by proposing a method for producing smooth estimators for the β_x's.

The article is organized as follows. Section 2 is devoted to the classical Lee–Carter method. The parameters α_x, β_x and κ_t are estimated by penalized least squares, with

a roughness penalty selected by cross validation. In Section 3, the same analysis is performed in the Poisson log-bilinear model. Numerical illustrations based on the French population are proposed. Our data are about French male population aged 0 to 98 between 1950 and 2000. The data related to calendar years 1950 to 1997 come from the *Institut National d'Etudes Démographiques* (INED) based in Paris, France. Those related to calendar years 1998, 1999 and 2000 have been obtained from the *Institut National de la Statistique et des Etudes Economiques* (INSEE) based in Paris, France. All the data are available from the Human Mortality Database (maintained by the University of California, Berkeley, USA, and the Max Planck Institute for Demographic Research, Germany), www.mortality.org or www.humanmortality.de (data downloaded in April 2006).

The effect of smoothing the β_x's on the mortality curves is investigated in Section 4. The final Section 5 concludes with a comparison with other possible approaches, including the P-spline method proposed by Durban *et al.* (2004) and Bayesian models worked out in Czado *et al.* (2005) and Girosi and King (in press). In this paper, it will be assumed that mortality statistics are available from some ages $x_{\min}$ to $x_{\max}$, and some calendar years $t_{\min}$ to $t_{\max}$. The data thus fill a rectangular array. This is however not necessary, and data with an irregular form are allowed. The results given below are easily extended to this more general setting.

2 Lee–Carter model

2.1 Penalized least squares

Let $\widehat{\mu}_x(t)$ be the empirical death rates at age x in year t (i.e. the ratio of the observed number of deaths D_{xt} to the corresponding exposure-to-risk er_{xt}). The estimated Lee–Carter parameters are traditionally obtained by minimizing $\sum_{x,t}(\ln \widehat{\mu}_x(t) - \alpha_x - \beta_x\kappa_t)^2$. This produces estimated β_x's and κ_t's with irregular shape in the majority of empirical studies. To smooth the estimated β_x's, we use here the objective function

$$\sum_{x=x_{\min}}^{x_{\max}} \sum_{t=t_{\min}}^{t_{\max}} (\ln \widehat{\mu}_x(t) - \alpha_x - \beta_x\kappa_t)^2 + \boldsymbol{\beta}' P_\beta \boldsymbol{\beta},$$

where

$$P_\beta = \pi_\beta \boldsymbol{\Delta}'\boldsymbol{\Delta} \quad \text{with } \boldsymbol{\Delta} = \begin{pmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & -2 & 1 \end{pmatrix}$$

and π_β is the smoothing parameter. The term $\boldsymbol{\beta}' P_\beta \boldsymbol{\beta}$ penalizes irregular β_x's. The objective function can therefore be seen as a compromise between goodness of fit (first term) and smoothness of the β_x's (second term). The penalty involves the sum of the squared second-order differences of the β_x's, that is, the sum of the square of the second differences $\beta_{x+2} - 2\beta_{x+1} + \beta_x$. Second-order differences penalize deviations from the linear trend. The trade-off between fidelity to the data (governed by the sum

of squared residuals) and smoothness (governed by the penalty term) is controlled by the smoothing parameters π_β. The larger the smoothing parameters the smoother the resulting fit. In the limit ($\pi_\beta \to \infty$) we obtain a linear fit. The choice of the smoothing parameters is crucial as we may obtain quite different fits by varying the smoothing parameters π_β. As explained below, the choice of the optimal π_β will be based on the observed data, using cross validation.

Eilers and Marx (1996) discuss discrete roughness penalties in detail. They show that a parametric polynomial fit of order $d - 1$ is approached when the penalty involves differences of order d and the weight of the penalty is increased. We refer the interested reader to this article for more details.

The estimated parameters can be obtained with the help of the following algorithm obtained from an application of Newton–Raphson methods: starting from some initial admissible values $\widehat{\boldsymbol{\alpha}}^{(0)}$, $\widehat{\boldsymbol{\beta}}^{(0)}$ and $\widehat{\boldsymbol{\kappa}}^{(0)}$ for the parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and $\boldsymbol{\kappa}$, we build the sequence of the $\widehat{\boldsymbol{\alpha}}^{(k)}$, $\widehat{\boldsymbol{\beta}}^{(k)}$ and $\widehat{\boldsymbol{\kappa}}^{(k)}$ from

$$\begin{aligned} C_\alpha^{(k)}\widehat{\boldsymbol{\alpha}}^{(k+1)} &= C_\alpha^{(k)}\widehat{\boldsymbol{\alpha}}^{(k)} + r_\alpha^{(k)} \\ C_\kappa^{(k)}\widehat{\boldsymbol{\kappa}}^{(k+1)} &= C_\kappa^{(k)}\widehat{\boldsymbol{\kappa}}^{(k)} + r_\kappa^{(k)} \\ (C_\beta^{(k)} + \mathrm{diag}P_\beta)\widehat{\boldsymbol{\beta}}^{(k+1)} &= (C_\beta^{(k)} + \mathrm{diag}P_\beta - P_\beta)\widehat{\boldsymbol{\beta}}^{(k)} + r_\beta^{(k)}. \end{aligned}$$

where $\mathrm{diag}P_\beta$ is the matrix with only the main diagonal of P_β, and $C_\alpha^{(k)}$, $C_\kappa^{(k)}$ and $C_\beta^{(k)}$ are diagonal matrices with dimensions $(x_{\max} - x_{\min} + 1) \times (x_{\max} - x_{\min} + 1)$, $(t_{\max} - t_{\min} + 1) \times (t_{\max} - t_{\min} + 1)$ and $(x_{\max} - x_{\min} + 1) \times (x_{\max} - x_{\min} + 1)$, and whose elements are

$$\begin{aligned} (C_\alpha^{(k)})_{x-x_{\min}+1,x-x_{\min}+1} &= (t_{\max} - t_{\min} + 1) \\ (C_\kappa^{(k)})_{t-t_{\min}+1,t-t_{\min}+1} &= \sum_{x=x_{\min}}^{x_{\max}} (\widehat{\beta}_x^{(k)})^2 \\ (C_\beta^{(k)})_{x-x_{\min}+1,x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} (\widehat{\kappa}_t^{(k+1)})^2 \end{aligned}$$

and $r_\alpha^{(k)}$, $r_\kappa^{(k)}$ and $r_\beta^{(k)}$ are column vectors with dimensions $(x_{\max} - x_{\min} + 1)$, $(t_{\max} - t_{\min} + 1)$ and $(x_{\max} - x_{\min} + 1)$, and whose elements are

$$\begin{aligned} (r_\alpha^{(k)})_{x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} (\ln \widehat{\mu}_x(t) - \widehat{\alpha}_x^{(k)} - \widehat{\beta}_x^{(k)}\widehat{\kappa}_t^{(k)}) \\ (r_\kappa^{(k)})_{t-t_{\min}+1} &= \sum_{x=x_{\min}}^{x_{\max}} \widehat{\beta}_x^{(k)}(\ln \widehat{\mu}_x(t) - \widehat{\alpha}_x^{(k+1)} - \widehat{\beta}_x^{(k)}\widehat{\kappa}_t^{(k)}) \\ (r_\beta^{(k)})_{x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} \widehat{\kappa}_t^{(k+1)}(\ln \widehat{\mu}_x(t) - \widehat{\alpha}_x^{(k+1)} - \widehat{\beta}_x^{(k)}\widehat{\kappa}_t^{(k+1)}). \end{aligned}$$

The convergence of the $\widehat{\boldsymbol{\alpha}}^{(k)}$'s, $\widehat{\boldsymbol{\beta}}^{(k)}$'s and $\widehat{\boldsymbol{\kappa}}^{(k)}$'s often requires just a few steps. Possible starting points are the estimations provided by ordinary least-squares (i.e. those obtained from the classical Lee–Carter method).

2.2 Selection of the smoothing parameter π_β via cross validation

The roughness penalty is controlled by the coefficient π_β, which has been assumed to be given in the preceding section. Here, we describe a method to select an optimal value of π_β from the data set (data-driven procedure). Let us define the prediction errors e_{xt,π_β} as

$$e_{xt,\pi_\beta} = \ln \widehat{\mu}_x(t) - \widehat{\alpha}^{-(x,t)}_{x,\pi_\beta} - \widehat{\beta}^{-(x,t)}_{x,\pi_\beta}\widehat{\kappa}^{-(x,t)}_{t,\pi_\beta} \tag{2.1}$$

where $\widehat{\alpha}^{-(x,t)}_{x,\pi_\beta}$, $\widehat{\beta}^{-(x,t)}_{x,\pi_\beta}$ and $\widehat{\kappa}^{-(x,t)}_{t,\pi_\beta}$ are the penalized least-squares estimators of the corresponding parameters, obtained with a roughness penalty equal to π_β on the basis of a data set without the observation for age x and year t.

The optimal value $\widehat{\pi}_\beta$ for the roughness penalty is then obtained from

$$\widehat{\pi}_\beta = \arg\min_{\pi_\beta} e^2_{\bullet\bullet,\pi_\beta}, \tag{2.2}$$

where

$$e^2_{\bullet\bullet,\pi_\beta} = \sum_{x=x_{\min}}^{x_{\max}} \sum_{t=t_{\min}}^{t_{\max}} e^2_{xt,\pi_\beta}.$$

Of course, this procedure is rather time consuming, and only a grid search is possible.

2.3 Numerical illustrations

The estimation of the α_x's is almost not affected by the roughness penalty. Considering Figure 1, we see that the $\widehat{\alpha}_x$'s are similar without penalty ($\pi_\beta = 0$) and for different values of π_β. Figure 2 shows the smoothing of the estimated β_x's resulting from increasing values of π_β. Note also that this operation influences the estimations of the κ_t's, as it can be clearly seen from Figure 3. This is not a surprise given that the product $\beta_x\kappa_t$ appears in the specification of the force of mortality $\mu_x(t)$. The impact on the estimated α_x's is almost negligible, but one can see some modifications in the estimated κ_t's (especially for males). This is the advantage of the approach compared with a naive smoothing of the Lee–Carter $\widehat{\beta}_x$: here, the $\widehat{\alpha}_x$ and $\widehat{\kappa}_t$ are adapted to the change in the $\widehat{\beta}_x$. The optimal smoothing coefficients are $\widehat{\pi}_\beta \approx 10^5$ for French males and $\widehat{\pi}_\beta \approx 10^6$ for French females. Note that the peak in the relative differences for the estimated κ_t's (bottom panel of Figure 3) is due to estimated values close to 0.

Figure 4 shows the fit of the forces of mortality for the year 2000, obtained with the roughness penalty. The smoothing effect is clearly visible, and the overall quality

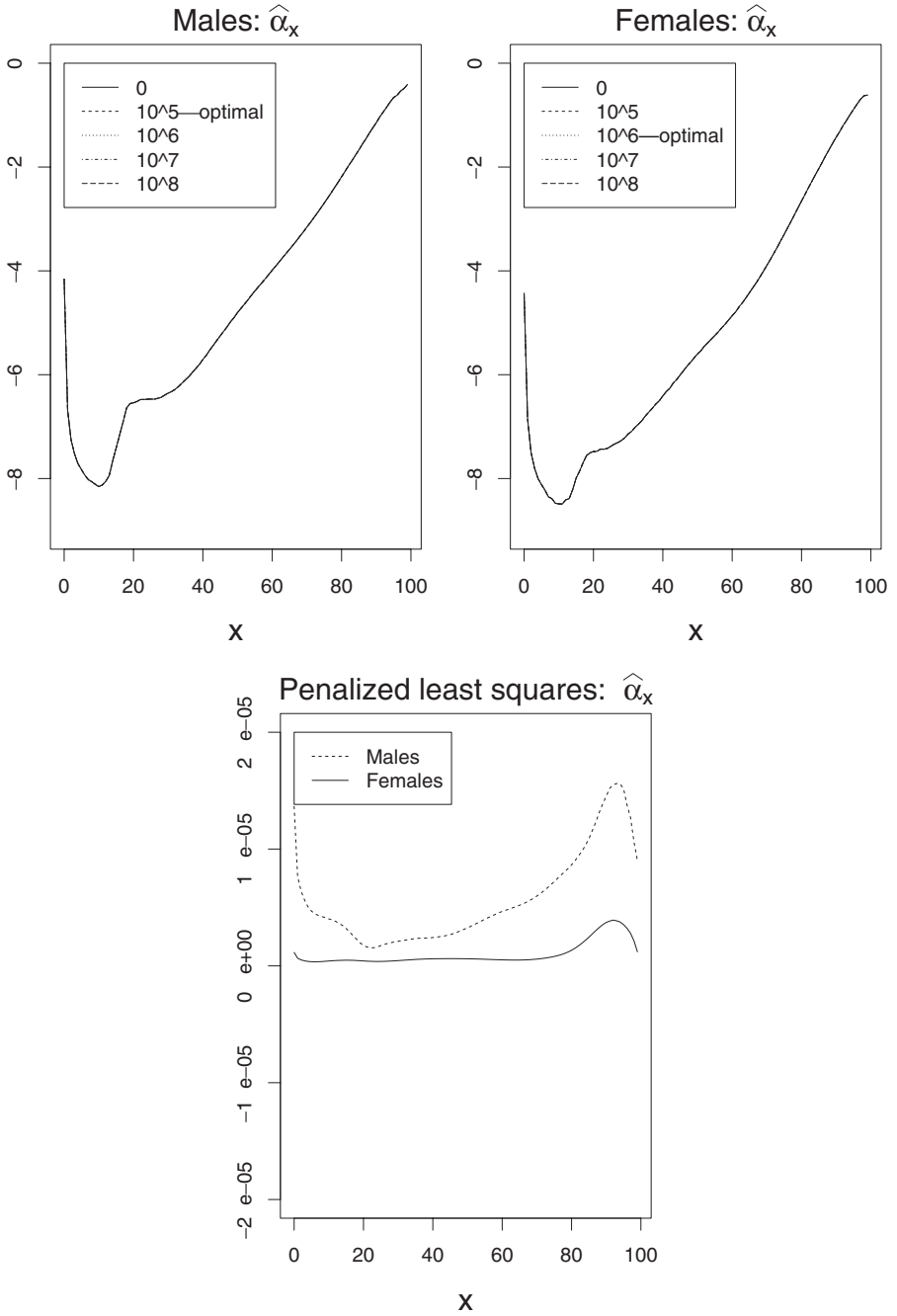


Figure 1 Estimation of the α_x parameters via penalized least squares for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated α_x's for males (top left), estimated α_x's for females (top right) and relative differences between the estimated α_x's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

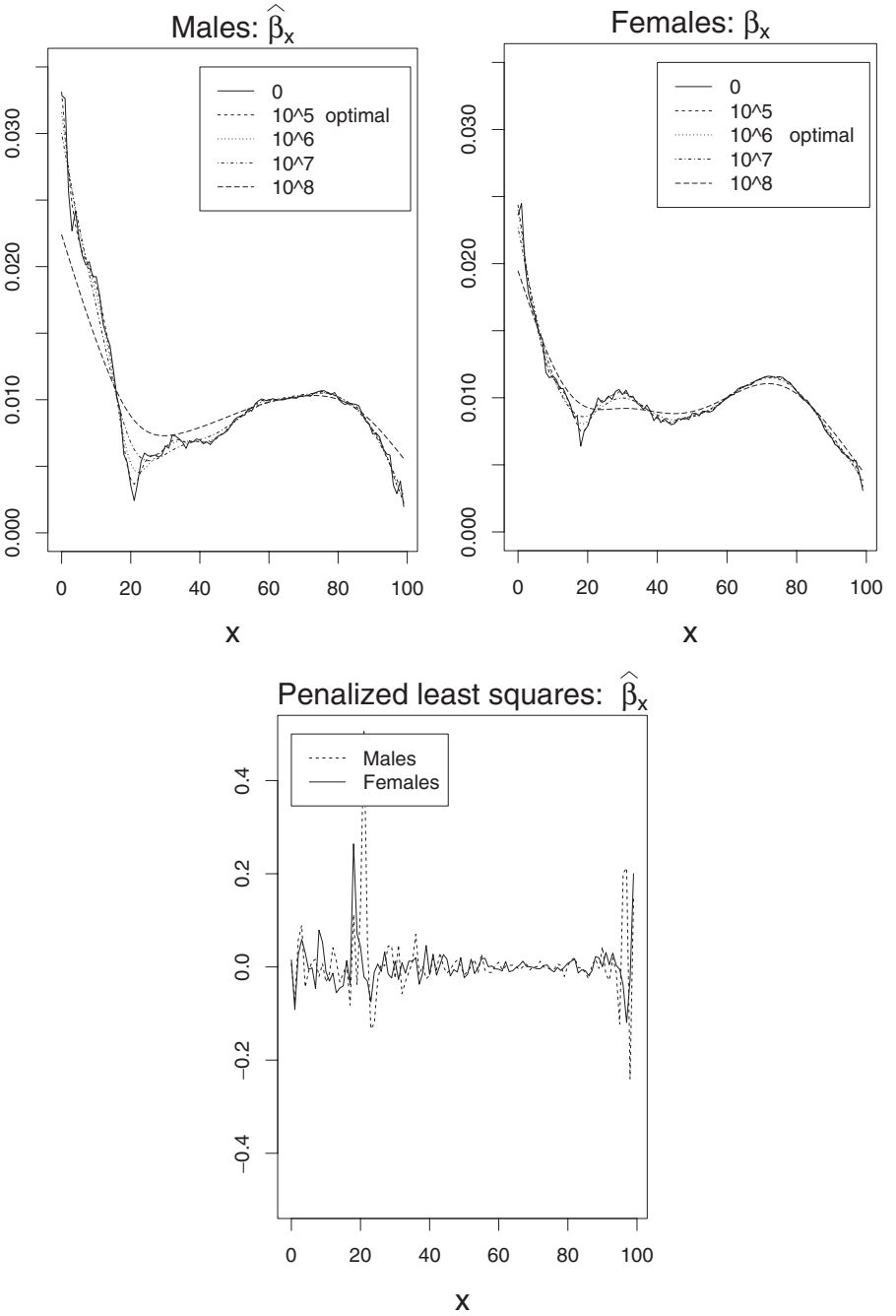


Figure 2 Estimation of the β_x parameters via penalized least squares for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated β_x's for males (top left), estimated β_x's for females (top right) and relative differences between the estimated β_x's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

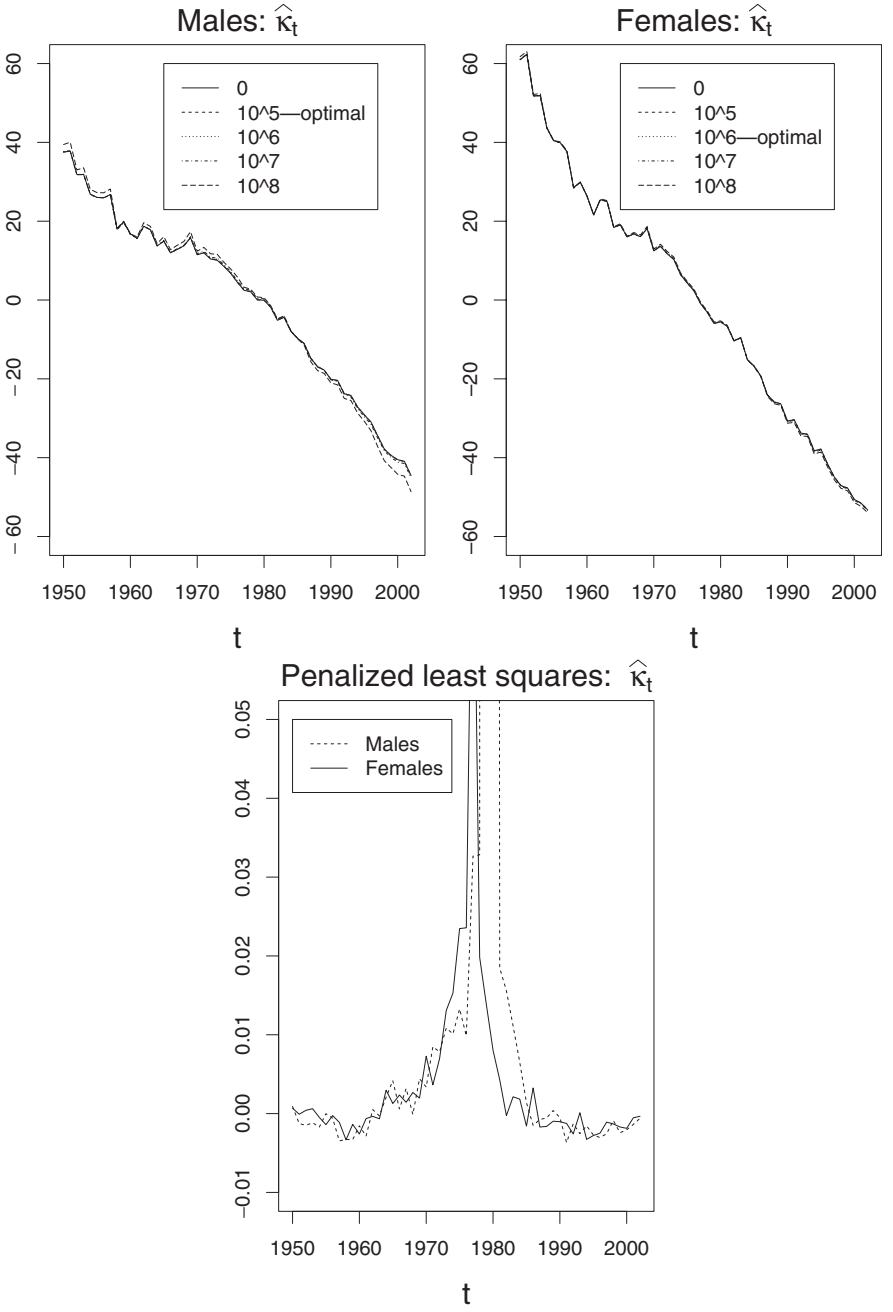


Figure 3 Estimation of the κ_t parameters via penalized least squares for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated κ_t's for males (top left), estimated κ_t's for females (top right) and relative differences between the estimated κ_t's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

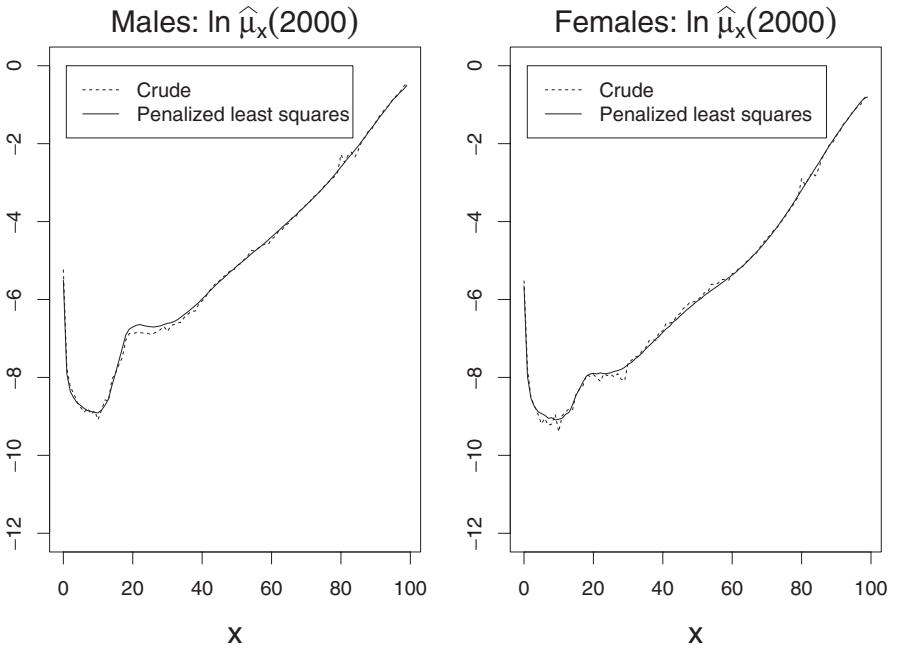


Figure 4 Observed and fitted forces of mortality for the year 2000: males and females

of the fit is very satisfactory. Note that the so-called 'accident hump' (high death rates originating from violent deaths around ages 18–30, mainly due to suicides and traffic accidents) is still apparent after smoothing (especially for males). This is a desired feature since this hump comes from a particular shape in the mortality curve, and is not due to sampling variations.

3 Poisson log-bilinear model

3.1 Penalized log likelihood

In the Poisson log-bilinear model, the number of deaths D_{xt} recorded at age x in calendar year t is assumed to be Poisson distributed, with mean $er_{xt}\mu_x(t)$ (recall that er_{xt} is the exposure-to-risk for age x and year t). As far as the force of mortality $\mu_x(t)$ is concerned, the same form $\exp(\alpha_x + \beta_x\kappa_t)$ as in the Lee–Carter model is kept in the Poisson model.

In this case, penalized least squares become penalized log likelihood. Specifically, denoting as $\mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\kappa})$ the likelihood associated to the Poisson model, that is

$$\ln \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\kappa}) = \sum_{x,t} \left\{ D_{xt}(\alpha_x + \beta_x\kappa_t) - er_{xt} \exp(\alpha_x + \beta_x\kappa_t) \right\} + \text{constant},$$

the estimations are obtained by maximizing

$$\ln \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\kappa}) - \frac{1}{2}\boldsymbol{\beta}' P_\beta \boldsymbol{\beta} = \ln \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\kappa}) - \frac{1}{2}\pi_\beta \boldsymbol{\beta}' \boldsymbol{\Delta}' \boldsymbol{\Delta} \boldsymbol{\beta}.$$

The estimated parameters can be obtained with the aid of the following algorithm obtained from an application of Newton–Raphson methods:

$$\begin{aligned} C_\alpha^{(k)} \widehat{\boldsymbol{\alpha}}^{(k+1)} &= C_\alpha^{(k)} \widehat{\boldsymbol{\alpha}}^{(k)} + r_\alpha^{(k)} \\ C_\kappa^{(k)} \widehat{\boldsymbol{\kappa}}^{(k+1)} &= C_\kappa^{(k)} \widehat{\boldsymbol{\kappa}}^{(k)} + r_\kappa^{(k)} \\ (C_\beta^{(k)} + diag P_\beta) \widehat{\boldsymbol{\beta}}^{(k+1)} &= (C_\beta^{(k)} + diag P_\beta - P_\beta) \widehat{\boldsymbol{\beta}}^{(k)} + r_\beta^{(k)}, \end{aligned}$$

where $C_\alpha^{(k)}$, $C_\kappa^{(k)}$ and $C_\beta^{(k)}$ are diagonal matrices with dimensions $(x_{\max} - x_{\min} + 1) \times (x_{\max} - x_{\min} + 1)$, $(t_{\max} - t_{\min} + 1) \times (t_{\max} - t_{\min} + 1)$ and $(x_{\max} - x_{\min} + 1) \times (x_{\max} - x_{\min} + 1)$, and whose elements are

$$\begin{aligned} (C_\alpha^{(k)})_{x-x_{\min}+1, x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} er_{xt} \exp(\widehat{\alpha}_x^{(k)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k)}) \\ (C_\kappa^{(k)})_{t-t_{\min}+1, t-t_{\min}+1} &= \sum_{x=x_{\min}}^{x_{\max}} (\widehat{\beta}_x^{(k)})^2 er_{xt} \exp(\widehat{\alpha}_x^{(k+1)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k)}) \\ (C_\beta^{(k)})_{x-x_{\min}+1, x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} (\widehat{\kappa}_t^{(k+1)})^2 er_{xt} \exp(\widehat{\alpha}_x^{(k+1)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k+1)}) \end{aligned}$$

and $r_\alpha^{(k)}$, $r_\kappa^{(k)}$ and $r_\beta^{(k)}$ are column vectors with dimensions $(x_{\max} - x_{\min} + 1)$, $(t_{\max} - t_{\min} + 1)$ and $(x_{\max} - x_{\min} + 1)$, and whose elements are

$$\begin{aligned} (r_\alpha^{(k)})_{x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} (d_{xt} - er_{xt} \exp(\widehat{\alpha}_x^{(k)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k)})) \\ (r_\kappa^{(k)})_{t-t_{\min}+1} &= \sum_{x=x_{\min}}^{x_{\max}} \widehat{\beta}_x^{(k)} (d_{xt} - er_{xt} \exp(\widehat{\alpha}_x^{(k+1)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k)})) \\ (r_\beta^{(k)})_{x-x_{\min}+1} &= \sum_{t=t_{\min}}^{t_{\max}} \widehat{\kappa}_t^{(k+1)} (d_{xt} - er_{xt} \exp(\widehat{\alpha}_x^{(k+1)} + \widehat{\beta}_x^{(k)} \widehat{\kappa}_t^{(k+1)})). \end{aligned}$$

3.2 Selection of the smoothing parameter by cross validation

As above, the selection of the optimal value for the roughness penalty coefficient π_β is based on cross validation. The prediction error e_{xt,π_β} is now given by

$$e_{xt,\pi_\beta} = \sqrt{2} \cdot \text{sign}(d_{xt} - \widehat{\delta}^{-(x,t)}_{xt,\pi_\beta}) \sqrt{d_{xt} \ln \frac{d_{xt}}{\widehat{\delta}^{-(x,t)}_{xt,\pi_\beta}} - (d_{xt} - \widehat{\delta}^{-(x,t)}_{xt,\pi_\beta})},$$

where

$$\widehat{\delta}^{-(x,t)}_{xt,\pi_\beta} = er_{xt} \exp(\widehat{\alpha}^{-(x,t)}_{x,\pi_\beta} + \widehat{\beta}^{-(x,t)}_{x,\pi_\beta} \widehat{\kappa}^{-(x,t)}_{t,\pi_\beta})$$

is the prediction of the average number of deaths, and $\widehat{\alpha}^{-(x,t)}_{x,\pi_\beta}$, $\widehat{\beta}^{-(x,t)}_{x,\pi_\beta}$ and $\widehat{\kappa}^{-(x,t)}_{t,\pi_\beta}$ are obtained by maximizing the penalized log likelihood with roughness penalty π_β omitting the observation for age x and calendar year t. Then,

$$\widehat{\pi}_\beta = \arg\min_{\pi_\beta} e^2_{\bullet\bullet,\pi_\beta} \text{ with } e^2_{\bullet\bullet,\pi_\beta} = \sum_{x=x_{\min}}^{x_{\max}} \sum_{t=t_{\min}}^{t_{\max}} e^2_{xt,\pi_\beta}.$$

3.3 Numerical illustrations

Figure 5 demonstrates that the $\widehat{\alpha}_x$'s are not influenced by the smoothness penalty as in the penalized least-squares approach. Figure 6 shows the smoothing of the $\widehat{\beta}_x$ resulting from increasing values of π_β. Note also that this operation influences the $\widehat{\kappa}_t$'s, as it can be clearly seen from Figure 7. The optimal smoothing coefficients are $\widehat{\pi}_\beta \approx 10^9$ for both French males and females. Again, the large values for the relative differences in the estimated κ_t's (bottom panel of Figure 7) come from estimated values close to 0.

Figure 8 shows the fit of the forces of mortality for the year 2000, obtained with the roughness penalty. The smoothing effect is again clearly visible, and the overall quality of the fit is satisfactory. The results are similar to those obtained by penalized least squares, and the remark about the accident hump still applies here. Differences between least squares and Poisson maximum likelihood eventually arise at older ages, where the exposures to risk are smaller (since these exposures are explicitly incorporated in the Poisson regression approach).

4 Effects on mortality curves

An important aspect of the methodology presented in this article is that the time factor $\widehat{\kappa}_t$ is intrinsically viewed as a stochastic process as initially proposed by Lee and Carter (1992). Box–Jenkins techniques are used to estimate and forecast κ_t within an ARIMA times series model. These forecasts in turn yield projected age-specific mortality rates, life expectancies and annuities single premiums. The random

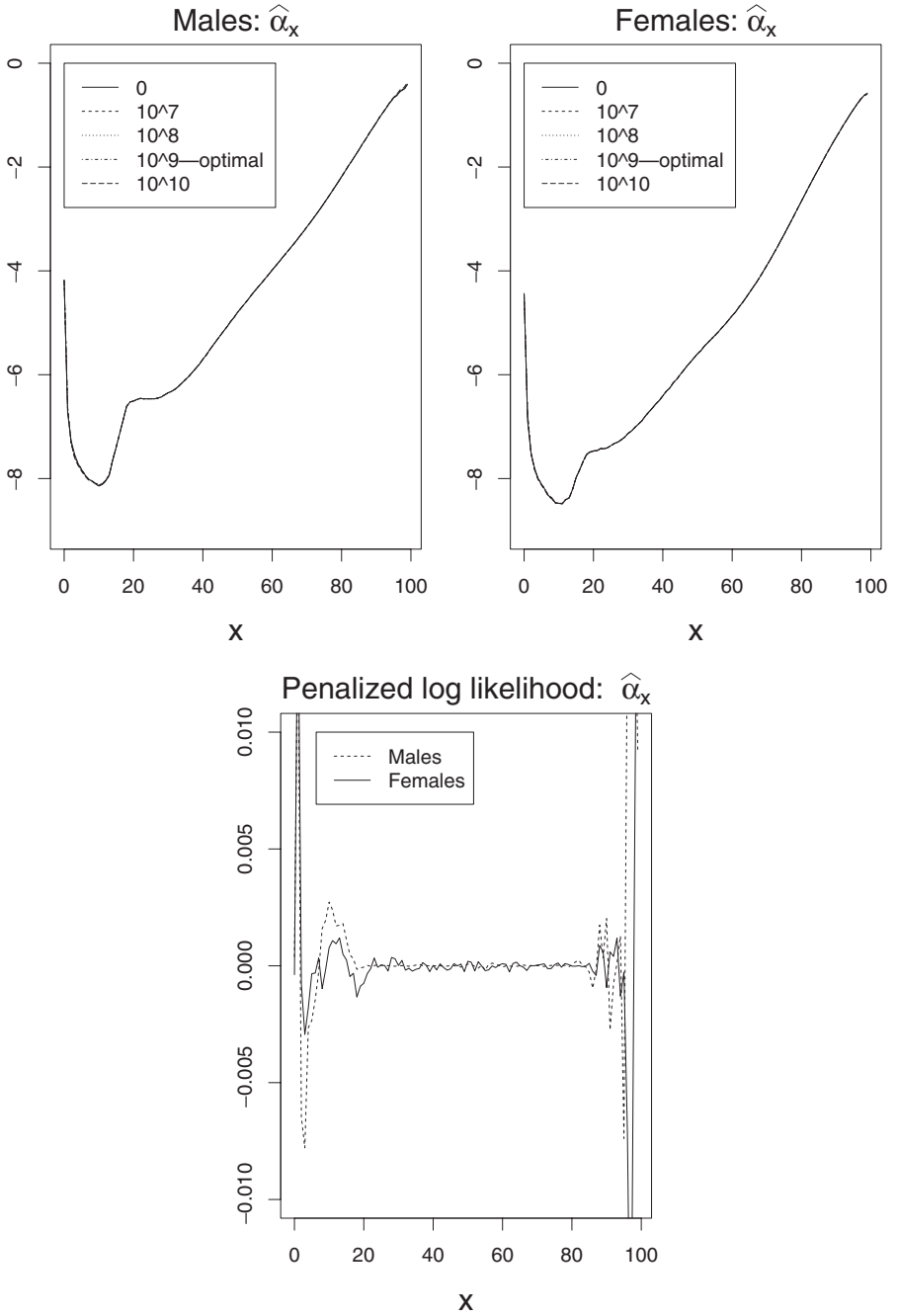


Figure 5 Estimation of the α_x parameters via penalized maximum likelihood for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated α_x's for males (top left), estimated α_x's for females (top right) and relative differences between the estimated α_x's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

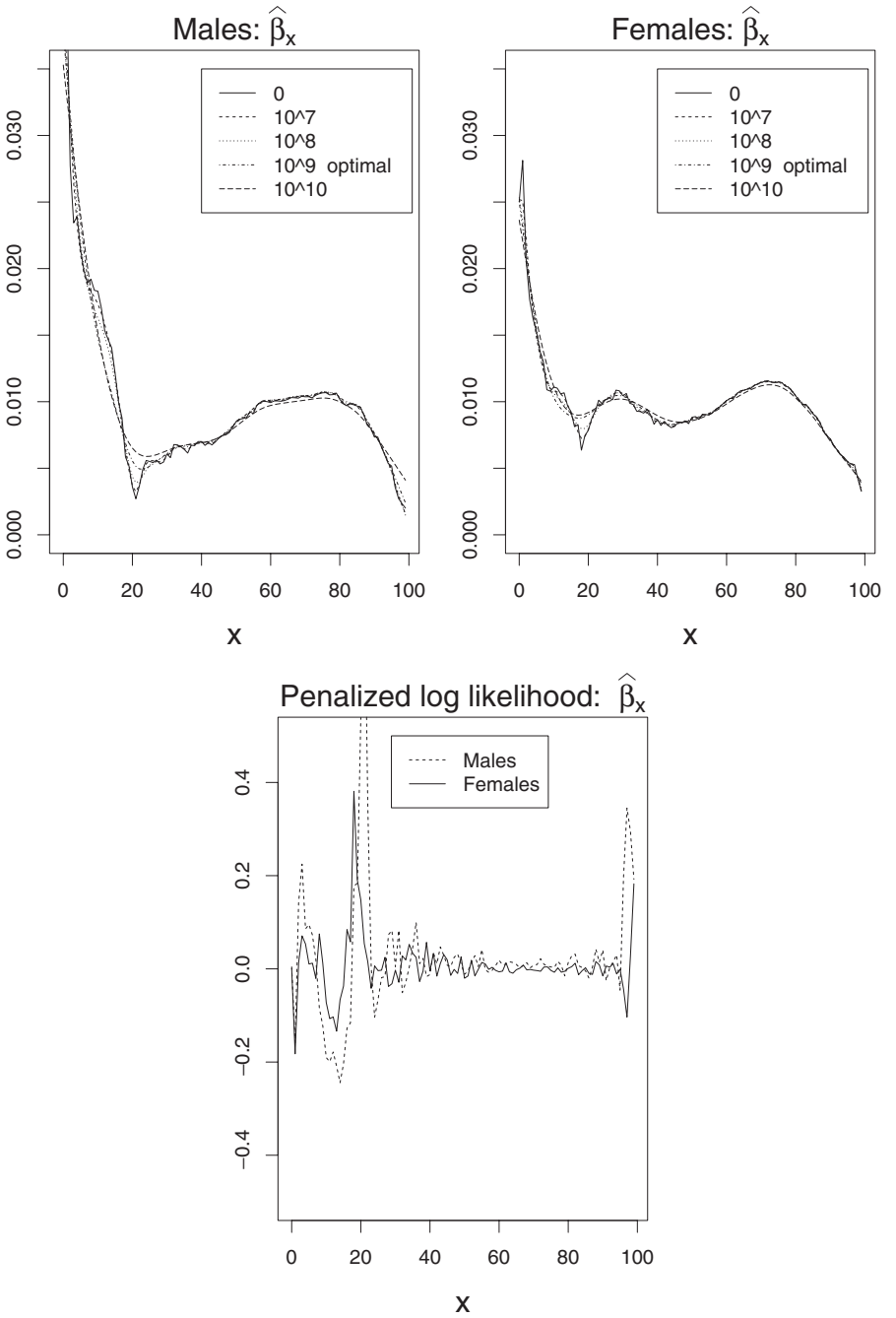


Figure 6 Estimation of the β_x parameters via penalized maximum likelihood for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated β_x's for males (top left), estimated β_x's for females (top right) and relative differences between the estimated β_x's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

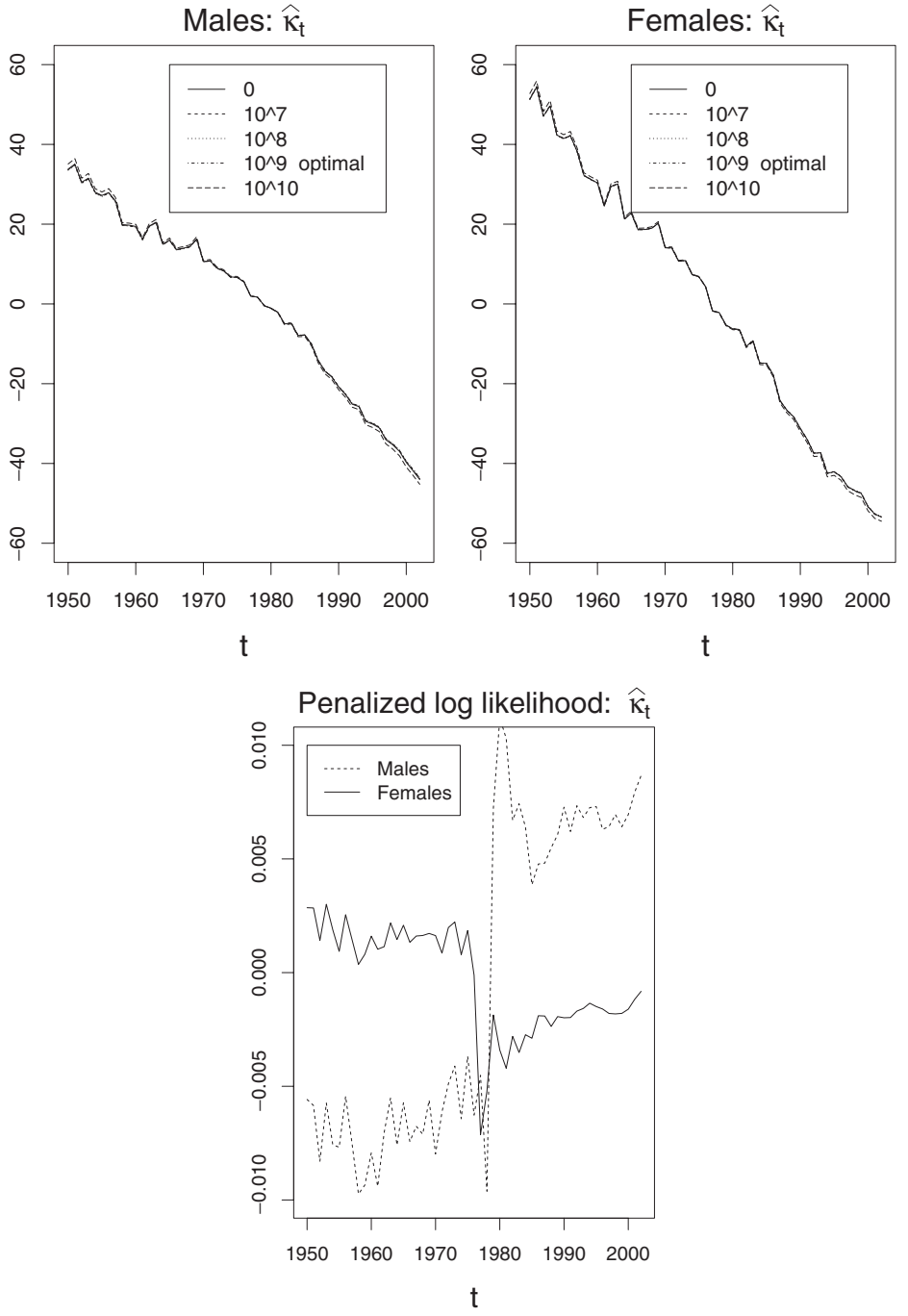


Figure 7 Estimation of the κ_t parameters via penalized maximum likelihood for different values of π_β including the optimal $\widehat{\pi}_\beta$ selected by cross validation: estimated κ_t's for males (top left), estimated κ_t's for females (top right) and relative differences between the estimated κ_t's obtained without smoothing ($\pi_\beta = 0$) and with the optimal smoothing coefficient $\widehat{\pi}_\beta$ (bottom)

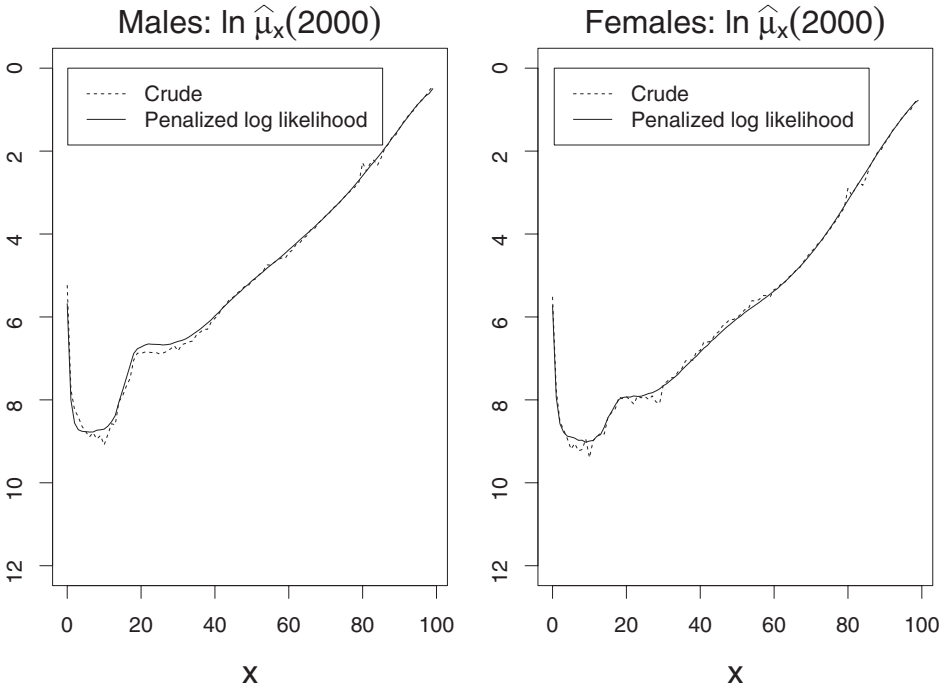


Figure 8 Observed and fitted forces of mortality for the year 2000: males and females

walk with drift (or ARIMA(0,1,0) process) features prominently in the published applications of the method proposed by Lee and Carter (1992). Here, we select the optimal ARIMA model for projecting the κ_t's to the future. The ARIMA(1,1,0) is selected for the κ_t's estimated by penalized Least squares, whereas the ARIMA(0,1,1) is preferred when the κ_t's are estimated by penalized Poisson likelihood. Specifically, the optimal models are of the form

$$(1 - B)\kappa_t = \mu + \frac{(1 - \theta_1 B)}{(1 - \phi_1 B)} a_t,$$

where B is the delay operator (i.e., $B\kappa_t = \kappa_{t-1}$), and where the a_t's are the random errors, centered with mean σ^2. The estimated ARIMA parameters are displayed in the following table, together with the respective minimal values of the AIC and BIC:

	μ	θ_1	ϕ_1	AIC	BIC
Pen. Least squares, Males	−1.57527	NA	−0.39598	220.03	223.93
Pen. Least squares, Females	−2.21465	NA	−0.43743	246.41	250.31
Pen. Log likelihood, Males	−1.4879	0.38367	NA	211.73	215.63
Pen. Log likelihood, Females	−2.05354	0.56673	NA	244.02	247.92

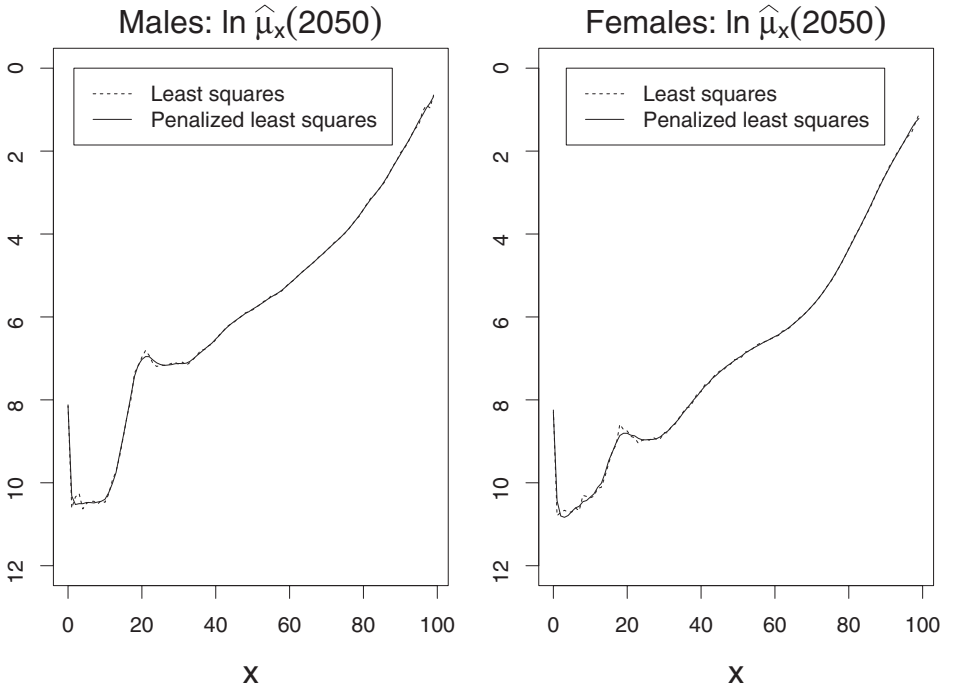


Figure 9 Forces of mortality for the French population in 2050 according to the smoothed Lee–Carter model for mortality projection (males are on the left, females are on the right)

Figures 9 and 10 make clear the effect of smoothing the β_x's on the resulting mortality curves. Specifically, the death rates that are likely to apply to the French population in 2050 are displayed in Figure 9, as a result of the Lee–Carter model for mortality projection. The effect of the smoothing is clearly visible: switching from the least-squares estimates to the penalized least-squares estimates produces a smoother curve. Figure 10 is the analogue for the Poisson log-bilinear model.

5 Conclusion

As mentioned in the introduction, the estimated β_x's obtained from least squares or maximum likelihood could have been smoothed leaving the estimated α_x's and κ_t's unchanged. This has been performed using the LOESS procedure of SAS/STAT (selecting the smoothing parameter by generalized cross validation, that is, by specifying the option select=gcv). Figure 11 shows the results. We observe relative differences up to 10–20%. Therefore, we cannot recommend the naive smoothing of the estimated β_x's.

Proper smoothing of the $\widehat{\beta}_x$'s with LOESS (or any other type of smoother) should account for different standard errors, leading to a relatively complex procedure. In contrast, the roughness penalty approach involves only a simple modification of the iterative re-weighted least-squares algorithm for Poisson regression. An additional

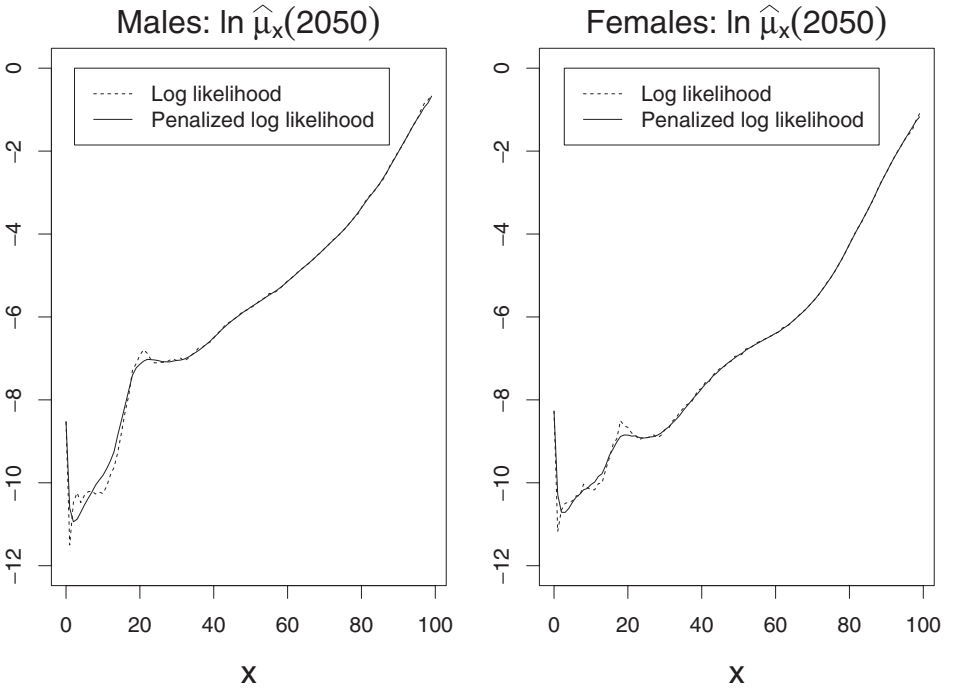


Figure 10 Forces of mortality for the French population in 2050 according to the smoothed Poisson log-bilinear model for mortality projection (males are on the left, females are on the right)

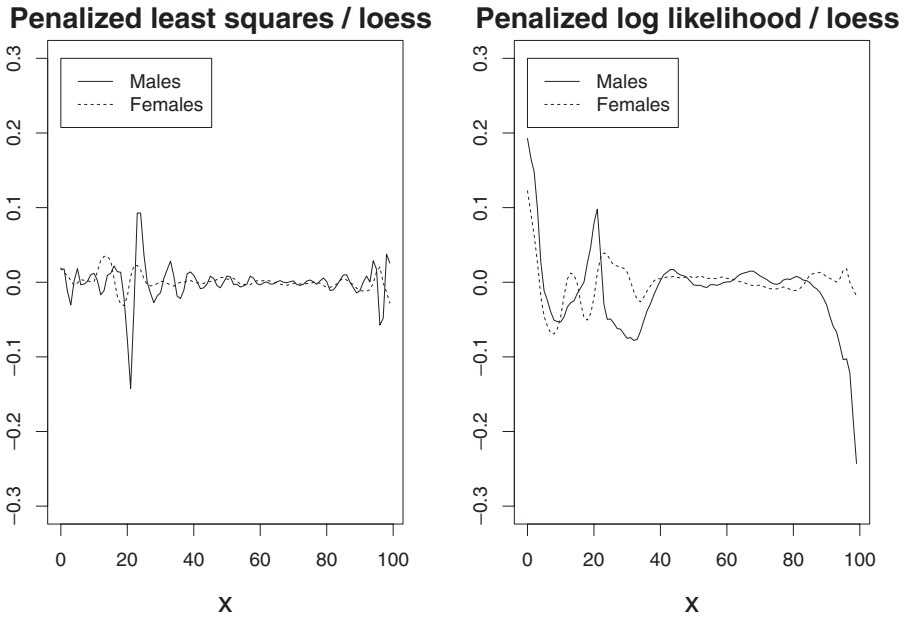


Figure 11 Relative differences of the smooth estimated β_x obtained from LOESS and penalized least squares (on the left) or penalized Poisson likelihood (on the right)

advantage is that an effective model dimension can be computed, which can be exploited to develop fast cross validation or the computation of AIC.

The alternative consisting in smoothing the resulting projected mortality curves (e.g., the series of the $\widehat{\mu}_x(2050)$) would destroy the decomposition of the forecast in the three series of parameters α_x, β_x and κ_t. Since this decomposition greatly simplifies the actuarial computations, this approach cannot be retained for insurance applications.

It is interesting to compare the methodology proposed in this article to some possible competitors. Czado *et al.* (2005) implemented a Bayesian version of the Poisson log-bilinear model for mortality projection. In this context, smoothness could be incorporated by an appropriate choice of a priori distributions. A Bayesian treatment of mortality projections has also been proposed by Girosi and King (in press). We refer to this book for alternative prior specifications imposing smoothness.

Another alternative to the approach described in this article is the method proposed in Durban *et al.* (2004). These authors smoothed death rates with P-splines in a Poisson model. In addition to the fact that these authors did not work within the Lee–Carter framework, an important difference is that forecasting in Durban *et al.* (2004) is a direct consequence of the smoothing process. The choice of the penalty then corresponds to a view of the future pattern of mortality.

Fitting the data and extrapolating past trends are kept separate in the present article (as in all the works in the vein of Lee and Carter (1992)). In the authors' opinion, this is an advantage for actuarial applications, since it allows for more flexibility. The degree of smoothing in empirical applications depends on the variability of the observed death rates. The size of the population under study, as well as the range of ages considered, thus both influence the smoothing coefficient and, possibly, the choice of the penalty. In the approach presented in this article, these features of the data are taken into account to estimate the time index, but do not interfere with the projection model (since it is a byproduct of the smoothing penalty).

Finally, let us also mention the recent work by Hyndman and Ullah (in press). These authors proposed a new method for forecasting age-specific mortality and fertility rates observed over time. Their approach extends the Lee–Carter model and allows for smooth functions of age, and is robust for outlying years due to wars and epidemics. Basically, the time-dependent function linking age to forces of mortality is decomposed in an appropriate functional basis, and the coefficients are projected to the future using ARIMA models. However, this approach is restricted to the modelling of death rates, whereas the present contribution also deals with observed death counts.

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